

Intrusion Detection System in IoT Network by using Metaheuristic Algorithm with Machine Learning Dimensional Reduction Technique

Chintam Anusha

Computer Science and Engineering
Vignan's Institute of Information
Technology

Visakhapatnam, India
anusha.rohini07@gmail.com

A. Sravani

Computer Science and Engineering
GITAM School of Technology

Visakhapatnam, India
sandavar@gitam.edu

Jetti Anusha

Computer Science and Engineering
GITAM School of Technology

Visakhapatnam, India
ajetti@gitam.edu3

Choudari Lakshmi

Computer Science and Engineering
GITAM School of Technology

Visakhapatnam, India
lchoudar@gitam.edu

Gantla Santoshi Kumari

Computer Science and Engineering
GITAM School of Technology

Visakhapatnam, India
sgantla@gitam.edu

Abstract— The network security is a main problem in any disseminated framework. To providing the safe and secure network we have been proposed anomaly detection system against from the suspicious attacks. The essential goal of this exploration is to plan effective IDS for IoT organization. The interruption discovery assumes a fundamental part in recognizing various assaults on IoT and upgrades the IoT performance. In this paper, one of the metaheuristic algorithms GSO (glow-worm swarm optimization) algorithm add with machine learning technique PCA (principal component analysis) used for providing the IDS. Therefore, the metaheuristic-based intrusion discovery model to recognize malicious activities by utilizing the NSL-KDD dataset. Dimensional reduction technique Principal component analysis is utilized for extraction of features and GSO algorithm is utilized to classify the various class of malicious attacks in the dataset. We have considered some metrics like precision, detection rate, recall, accuracy and FAR are computed. The suggested method evaluated better in each parameter compared with other existing techniques.

Keywords: *IoT, metaheuristic, Intrusion Detection System, Principal Component Analysis, NSL-KDD dataset, Glow-worm Swarm Optimization.*

I. INTRODUCTION

The RFID (Radio Frequency Identification Development) group first proposed the Internet of Things (IoT) in 1999, and it has since emerged as crucial to the real world's advancement of phones, implanted and all-encompassing communication, distributed computing, and information research [1]. The IoT is predicted to play a main role in bringing all needs of our daily lives together. It includes a variety of areas such as vehicles, health care, sports, entertainment, and upscale houses. The widespread use of IoT improves some daily activities, expands social interactions with people and objects, and advances how people interact with the environment [2].

The IoT is motivated by the idea of connecting more than only people and computers as commonplace objects to the network of network. This was accomplished by setting up things with registration and correspondence restrictions and then generally planning the real world to the contemporary one [3]. The underlying long periods of the IoT mostly involved M2M interaction for information exchange. However, the idea of fusing human correspondence with the Internet has evolved quickly [4]. However, comes at the expense of protection and security issues: if private, modified information is admitted to an unauthorised, malicious administrator, it may result in basic harm to our wealth, position, and personal security. Additionally, these devices incorporate components that their creators introduced at various points throughout the creation appropriation chain, according to our own unique data [5]. Wires, firmware, and investigation modes are some of these.

This research proposes a glow-worm swarm streamlining (GSO) technique with PCA for detection malicious on IoT networks. The majority of the time, normal GSO is used for image optimization and classification [6]. It is used in this study to identify anomalies in IoT enterprises. To extract features from the dataset, PCA is used. But in real situations, each factor weighs in differently on the decision. Due to this reality, our work first used PCA to extract the dataset's key features in order to reduce the information measured and shorten the time needed for classification.

The goal of this work is to construct an intrusion recognition model for identifying malicious and unexpected operations in IoT businesses. So, the suggested model uses the NSL-KDD dataset to distinguish between assaults and is based on metaheuristic calculation. The PCA-dependent GSO approach is used to play out the irregularity recognition process. The PCA is used to filter features, while the GSO approach is used to classify objects.

II. RELATED WORK

An abnormality discernment model for IoT network traffic utilising the PCA technique was proposed by Dang Hai Hoang and Ha Duong Nguyen. For the purpose of lowering larger data dimensions, PCA was applied. To deduce formulas from earlier research, a brand-new distance formula was put forth and implemented. These derivations served as the foundation for the implementation of a novel technique for anomaly identification in network traffic, which reduced computing cost while producing adequate findings utilising a new distance formula.

Dang et al. [7] Used an unsupervised model CNN and autoencoder that allowed for anomaly recognition of network traffic. To automatically profile traffic patterns and filter out abnormal traffic, the D-PACK abnormal traffic detection technique was utilized. The D-PACK, however, only examined the first few bytes of the first few Mirai-CCU and USTC-TFC datasets were the used for analysis. At last, the model significantly decreases the volume of traffic for detection with 100% veracity and less than 1% false negative and positive rates by reviewing fewer packets and total bytes from each packet possible.

Kshirsagar et al. [8] A feature reduction methodology was presented by using the J48 machine learning algorithm. It combines the correlation (CR) and information gain (IG) feature selection strategies. In order to effectively discern DDoS anomalies based on reflection and double-dealing, this review provides a framework for identifying DDoS intrusions.

Nimbalkar et al. [9] Gain Ratio (GR) and Information Gain (IG) are proposed features for anomaly detection systems employing JRip classifier by. This system obtains subset of features by applying union and insertion operations to get the ranked 50% GR and IG features and the system given the higher performance than the original features data set.

Chen et al. [10] Given The anomaly detection model using autoencoder model. Additionally, the dimensionality reduction in this case was accomplished using convolutional autoencoder (CAE). As is well known, compared to a regular autoencoder, this CAE requires less training time. This model can find irregular co-relations between features to increase detection precision. For this study, the National Security Lab-Knowledge Discovery data set was used, and the CAE model outperformed it in terms of identification.

Yansenz et al. [11] Suggested Multilevel Auto regression model for intrusion detection of network traffic model. The primary objective of this autoregression model was to detect DDOS attacks. Information entropy and multilayer autoregression models were applied to increase the network traffic detection rate. The entropy of the information network traffic per unit of time was initially computed by the model, and its time series were obtained using zero-mean. The entropy of the sequence was then predicted using a multilevel autoregression approach. The residual was then divided by the average value between the true and predicted values of entropy to see if an anomaly was present. Use this model to identify unknown anomalous traffic.

Maryam A et al. [12] Presented Metaheuristic algorithms like genetic, particle swarm, and glow-worm optimization were employed for IDS. These algorithms were subjected to

DDoS and DoS attacks, and the WSN nodes' lifetimes were evaluated. The PSO method proved superior in terms of permissivity, whereas the genetic method performed better in terms of energy usage. The GSO method has poor permissivity performance and increased energy consumption.

Ren-Hung Hwang et al. [13] used DL model that includes both autoencoder and CNN to recognize early anomalies in network data. For auto-profiling traffic patterns and aberrant traffic filtering, the D-PACK anomaly traffic detection technique was used. However, the D-PACK only looked at the first few bytes of the first few packets in each flow to look for early detection. The datasets utilized for the tests were USTC-TFC and Mirai-CCU. Finally, the model significantly reduced the traffic volume with 97% accuracy and less than 2 % false rates by looking at fewer packets and total bytes from each packet feasible.

Summary of related intrusion detection systems are shown in below Table I.

TABLE I. SUMMARY OF RELATED INTRUSION DETECTION SYSTEMS

Source	Category	Method	Dataset	Performance
Dang et al (2018). [7]	Intrusion detection system	PCA	Kyoto 2006	93.81TPR &5% FPR metrics
Kshirsagar et al (2021). [8]	DDoS	J48 classifier	KDD cup 1999 & CICDoS 2019	Reduction by 60.97 of original features
Nimbalkar et al (2021). [9]	Intrusion detection system	Jrip classifier	KDD Cup 1999	99.99% accuracy and 99.57% DR
Chen et al (2018). [10]	Anomaly detection	Convolutional autoencoder	NSL-KDD	96.58 accuracy
Yansenz, et al (2019). [11]	Anomaly detection of network traffic	Multilevel auto regression	Data packets	95% DR
Ren et al (2020). [13]	Network traffic anomaly detection	CNN	KDDCup 99 &NSL-KDD	93.8% TPR & 5%FPR metrics

III. METHODOLOGY

Every distributed system has a serious difficulty with network security. In order to safeguard the network against anomalous activity, IDS have been utilized. IDS is used to assist in identifying illegal intrusion scenarios by placing a security layer over the networks. The two primary IDS detection methods are intrusion-based and signature-based. The majority of IDS employ detection rules and are signature-based systems. However, a network is a large then it will require a lot of policies for IDS, it can be expensive and uses more time. IDS also makes use of attack signatures to spot malicious behavior. Intruders might be able to enter the network if the signatures were not sufficiently defined. Systems based on anomalies don't require the intervention of humans that has been proposed as a solution to these issues. The working processes of the suggested model is given in Fig.1.

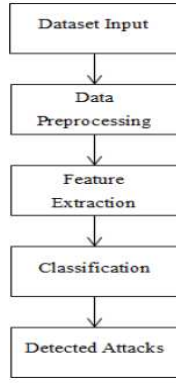


Fig.1. Flow diagram of suggested model

The two types of IDS that are based on the concept of interruption location are Network-based ID'S (NIDS) and Host-based ID'S (HIDS). For some IoT device applications with restricted resources and processes, the HIDS is inappropriate. The network ID'S can observe all network outbound congestion and identify both well-known and obscure intrusions that used an IoT process with both authorized and abnormality-based techniques.

Generally speaking, the inconsistency-based NIDS approaches are essential for tracking organization traffic and differentiating fresh attacks. The mark-based NIDS can't be used to identify one attack from another in upcoming business transactions. In light of this, a GSO computation with PCA is used to develop the suggested model, which is designed to recognize irregularity-based NIDS-based assaults.

A. Dimension Reduction Technique

Principal component analysis is a method for measurement decrease, it changes over a bunch of genuine correlative factors into a gathering of uncorrelated factors, in particular Principal Components (PC). These PCs are the first factor's direct mixes. The all-out PCs got are more modest than the absolute genuine factors, or equivalent. PCA along these lines empowers for a lower intricacy. It was considered as straightforward yet proficient method for the identification of organization-based anomalies. The basic idea of principal component analysis was to make a direct change sequence for the genuine information it has explicit related attributes. Consequently, the key segment load network modified into the gathering of new information with least ascribes to mean the genuine information. It is fitting for the interaction of N-dimensional information reduction.

Step1: Represent the perception lattice Y for genuine information. The perception for x factors t_1, t_2, t_x decide network Y. Each line portrays mathematical assessment of test information of the dataset; segment number n views n tests were found according to below Equation (1).

$$Y = \begin{bmatrix} y_{11} & \cdots & y_{1x} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{nx} \end{bmatrix} \quad (1)$$

Step2: For the perception grid, concentrate on processing of the data. Register both the standard deviation and mean is given in below equations (2) and (3).

$$\sim y_b = \frac{1}{n} \sum_{a=1}^n y_{ab} \quad (2)$$

Based on the formulas (3) and (4)

$$S_b = \sqrt{\frac{(y_{ab} - \sim y_b)^2}{n}} \quad (3)$$

$$\sim y_{ab} = \frac{y_{ab} - \sim y_b}{S_b} \quad (a, b = 1, 2, \dots, n) \quad (4)$$

Step3: Calculate the sample correlation matrix based on the below given Equation (5).

$$M = \frac{1}{n} \sim y^T \sim y \quad (5)$$

The evaluation of each value in W is as per below Equation (6)

$$M_{ab} = \frac{\sum_{k=1}^n (y_{ka} - \sim y_a)(y_{kb} - \sim y_b)}{\sqrt{\sum_{k=1}^n (y_{ka} - \sim y_a)^2 \sum_{k=1}^n (y_{kb} - \sim y_b)^2}} \quad (6)$$

Step4: Determine M's eigenvalue and eigenvector. Obtain M's x characteristic values, which are 1 through 2, etc. So, using the formula, determine each primary component's rate of contribution.

$$R_a = \frac{ra}{r_1 + r_2 + \dots + r_x} \quad (a=1, 2, \dots, x) \quad (7)$$

and select the largest PC that contains 83% and include $\lambda_{p+1} < 1$ as PCA outputs. Calculate the corresponding Eigen vectors $e_1, e_2, \dots, e_x$. and select the top p variables for the PCL. Decreasing order characteristics values are 1, 2... x.

$$C_{x \times p} = (e_1, e_2, \dots, e_p) \quad (8)$$

Step5: Create straight change for the genuine information of PCL lattice $C_{x \times p}$ below Equation, to make new factors of head part $m_1, m_2, \dots, m_p$.

$$\begin{bmatrix} m_1 \\ \vdots \\ m_2 \end{bmatrix} = C_{x \times p}^T \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_2 \end{bmatrix} \quad (9)$$

The network measurement diminished from x top after the direct change and consequently, fundamentally diminishes the example information volume.

B. GSO (Glowworm Swarm Optimization)

The Glow-worm Swarm Optimization was developed using the idea of glow-worms' foraging or courting behavior, which glows to attract companions. The amount of luciferin in each glow-worm in GSO controls the luminance intensity. Every glow-worm in an algorithmic procedure goes toward its neighbor who was brighter than it. The swarm can split into several subgroups through these activities, which were only dependent on local data, and merge to perform multimodal functions through various optimum. GSO contains four main stages: distribution of glow-worms, updating of luciferin, movement of glow-worms, and updating of neighborhood ranges [13].

1) Glowworm Distribution

The n glow-worms are randomly scattered over the search space at different locations. The amount of luciferin l_0 carried by each glow-worm is the same.

2) Luciferin Updation

The worth of previous individual luciferin levels and goal function is the foundation of this stage. The luciferin update policy was

$$B_p(j) = (1-\rho)^{b_{p-1}(j-1)} + \gamma G(y_p(j)) \quad (10)$$

where, $G(y_p(i))$ is the final function value at the glow-worm g 's place $B_p(i)$ given the glow-worm p 's luciferin value at the j^{th} step, ρ and γ were the decay and luciferin enhancement factor.

3) Glow-worm Movement Stage

At this step, every glow-worm draws close to a neighbor who has a higher brightness than themselves using a probabilistic method. The likelihood of a glow-worm g migrating toward a neighbor was

$$P_{ph}(i) = \frac{Lh(i) - Lp(i)}{\sum_{j \in n_p(i)} (Lj(i) - Lp(i))} \quad (11)$$

here, $h \in n^{\wedge}(i)$, $n_p(i)$ is a set, which could be validate by Equation, (12).

$$n_p(i) = \{h: d_{ph}(i) < r_d^p(i); l_p(i) < l_h(i)\} \quad (12)$$

here $d_{ph}(i)$ gives the distance between glowworms p and h at the i^{th} step and $r_d^p(i)$ gives the variable neighborhood range related with glowworm p at the i^{th} step.

4) Neighbourhood Region Updation

The range of neighborhood each glow-worm was shown in the generation, using r_0 as the first neighborhood domain:

$$X_{d^p}(i+1) = \min \{x_s, \max \{0, x_{d^p}(i) + \beta(ni - Np(i))\}\} \quad (13)$$

where x_d stands for the sensory radius of the glow-worm, ni was a number of neighborhoods control attribute, was a constant, and $Ng(i)$ stands for neighborhoods set.

In this paper, Ordinarily GSO calculation is essentially utilized for picture order and streamlining. Here utilized for interruption discovery in iot organizations. For highlight filtering of dataset, the Principal Component Analysis is utilized. Notwithstanding, in cases of genuinely, each element diversely influences the choice. Because of this reality, this work at first separated essential highlights from the dataset utilizing PCA, for diminishing the information measurement, and decreasing the span of characterization.

IV. RESULTS AND DISCUSSIONS

The proposed model implementation and assessment by using MATLAB. The suggested method will be evaluated using the outcome factors precision, recall, FAR, accuracy and detection rate. For the suggested GSO-PCA methodology, the analysis will be contrasted with other methods including PSO, SVM, ANN and BPNN.

A. Dataset Description

To overcome KDD99's limitations, Data-mining (NSL-KDD) dataset and the National Security Lab- Knowledge Discovery was enhanced. The public can access the dataset through its official website. Duplicate records were initially eliminated from the training and test sets. Then, a more number of records are chosen from the KDD99 in order to produce precise findings for classification systems. Thirdly, it resolved the issue of an uneven probability distribution. Table II. Represents with 36874 training examples and 23634 test cases in the NSL- KDD data collection, with 38 consistent ,41 characteristics and 3 classes (discrete-valued). Six continuous variables were disregarded because they had significant 0s. The evaluated data set contains 39 groups, suggesting that there was no malicious into the evaluated data during pre-processing. There are a total of 21 classes, including 23 training classes, 38 testing classes, and 17 classes that are unusual for test knowledge. Classes that are good for the evaluated dataset but are absent in the time pre-processing make up about 16.6% of the test data set samples. The classification is made more challenging by this distribution variance of the class.

There are five different classes that can be linked to the training/testing courses: Normal, PROBE, R2L, U2R, and DoS. With the exception of the Not attack class, each categorization is identical to an incursion that implies no malicious have been shown. These categories continue to be valuable in the context of ID's, are still largely unbalanced, and contain a sufficient number of instances in each class to produce more significant results. When a hacker uses host and network scanning techniques to try to obtain information from the target network, this is known as a probe (i.e., ports scanning).

DoS: When a hacker prevents legitimate users from accessing a service or system.

U2R: when an unauthorized user tries to get root access by extending a minimum user's advantage.

R2L: When a hacker remotely accesses a target computer by impersonating active local users. The U2R and R2L attacks were among the most heinous to be categorized as imitating typical user actions.

TABLE II. TRAFFIC DISTRIBUTION OF NSL-KDD

Traffic	Training	Test
DoS	45927	7458
Normal	68363	9821
Probe	12466	2645
U2R	52	200
R2L	985	2341
Total	136874	23634

Based on the aforementioned criteria and the dataset's identification of the attacks, the suggested model is assessed. DR denotes the classifier's attack detection rate; FAR denotes the percentage of false positives; and recall denotes the model's attack returns. Correct identification is the accuracy measure in each case. The accuracy establishes if the assaults that were returned were valid. It is necessary to assess the effectiveness of several outcome indicators in order to

evaluate the Glowworm optimization-principal component model and to contrast it with other contemporary.

TABLE III. THE ANALYSIS ASSESSMENT OF THE PROPOSED METHOD USED WITH THE DATASET

Attack	ACC	FAR	Precision	Recall
DoS	93.12	0.8	95.32	96.61
Probe	95.36	1.2	92.57	94.55
R2L	91.21	7.3	88.32	90.22
U2R	93.72	2.6	91.40	94.69

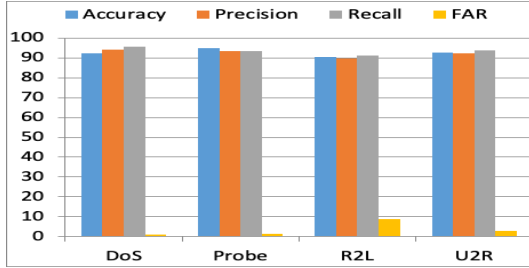


Fig.2. Performance analysis of proposed model

The Evaluation assessment of the proposed method used with the National Security Lab -Knowledge Discovery dataset is shown in Table III. where each assault class is evaluated based on its precision, accuracy, FAR and recall. The plot of the suggested model performance is shown in Fig. 2.

As previously stated, the suggested model's performance analysis was contrasted with those of other models, including PSO, support vector machines (SVM), artificial neural networks (ANN) and back-propagation neural networks (BPNN) as given in Table IV. Following the graphical display in Fig.3. classification accuracy is compared and validated here.

TABLE IV. CLASSIFICATION ACCURACY COMPARISON OF DIFFERENT ALGORITHMS

Method	Normal	DoS	R2L	Probe	U2R
ANN	91.20	88.53	81.77	89.20	66.02
SVM	90.72	86.35	89.11	87.85	79.75
BPNN	94.24	90.95	91.89	94.08	85.08
PSO	93.25	95.70	91.63	94.37	88.61
GSO-PCA	95.24	94.92	95.25	94.80	89.32

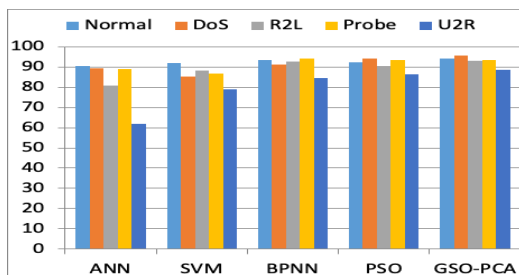


Fig.3. Comparison of different model's accuracy classification

When compared to the other methodologies, the suggested GSO-PCA performed good in the majority of performance metrics. For the class normal, the suggested method got 1 to 4.8% higher detection, for the DoS class, 3.36 to 13.72 % more detection, for the class R2L, 0.43 to 6.95%

more detection, for the probe class and for the class U2R, 1.3 to 23.3% higher detection obtained.

TABLE V. COMPARISON BETWEEN RATE OF DETECTION AND FALSE ACCEPTANCE RATE

Method	Rate of Detection (%)	False Acceptance Rate (%)
ANN	87.61	5.98
SVM	89.80	5.02
BPNN	92.82	3.95
PSO	92.98	3.07
GSO-PCA	95.12	2.97

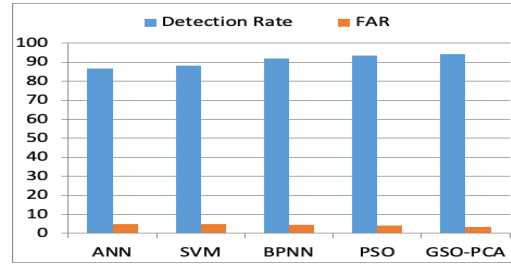


Fig.4. Comparison between detection rate and false acceptance rate

Table V. evaluates the Differentiation of Detection Rate and False Acceptance Rate, and Fig.4. depicts the plot of the differentiation, respectively. When compared to previous approaches, the suggested model had a higher detection rate and FAR rate. In terms of both the rate of detection and False acceptance rate, the Particle swarm optimization evaluated reasonably near to the outcomes of the suggested method. The proposed model had the maximum performance with 95.08% DR and 3.41% FAR, while ANN had the lowest performance with 87.52% DR and 6.06% FAR. The GSO-PCA model has produced detection rates that are 0.4 to 7.5% higher and FARs that are 0.5 to 1.6% lower.

In this section we also present that the comparison between our GSO-PCA intrusion detection system with JRip classifier from the study [11] and multilevel auto regression from the study [13] intrusion detection systems. The JRip classifier ID system having the detection rate (DR) is 99.57% but got with more misclassifications and multilevel autoregression detection system having detection rate is 95% with more information entropy. in our detection system we got Detection rate is 95.08 with less false classifications because we applied both routing and machine learning classification algorithms.

V. CONCLUSION

In this paper, it was suggested to use the GSO (glow-worm swarm optimization) algorithm and PCA (principal component analysis) to detect intrusions in networks. Generally speaking, picture categorization and optimization were the main uses of the GSO method. It was utilized in this study for IoT network intrusion detection. The dataset's feature extraction method was carried out using PCA. The proposed model was created in order to improve performance when spotting anomalies in IoT networks. The National Security Lab -Knowledge Discovery dataset was employed for the classification and detection of attacks. The PCA successfully extracts the attributes of the data set, and the Glow warm optimization is utilized to identify and classify the various types of malicious that are present in the data. Numerous performance attributes, including recall, precision, accuracy, FAR, and rate of detection, are assessed in order to

assess performance. Although not the best performance overall, the suggested model outperformed the other present approaches like ANN, BPNN SVM and PSO in every feature. In the further work, the suggested model may be used with various datasets in various network platforms, such as WSN, MANET, etc., and a hybrid model employing deep learning technique can be created to increase the rate of detection and decrease the FAR.

REFERENCES

- [1] A.R. Arif et al., "Intrusion detection system using hybrid binary PSO and K-nearest neighbourhood algorithm," *Proceedings of 11th International Conference on Information and Communication Technology and Systems, ICTS*, 2017.
- [2] M. Reazul, A. Rahman and T. Samad, "A network intrusion detection framework based on bayesian network using wrapper approach," *International Journal of Computers Applications*, vol.166, no.4, pp.13-14, 2017.
- [3] M.N. Aziz and T. Ahmad, "Cluster analysis-based approach features selection on machine learning for detecting intrusion," *International Journal of Intelligent Engineering & Systems*, vol.12, no.4, pp.233-243, 2019.
- [4] J. P. Papa and R. Munoz, "Internet of things: a survey on machine learning-based intrusion detection approaches," *Computer Networks*, vol. 151, pp. 147-157, 2019.
- [5] K. Khan, A. Mehmood, S. Khan, M. Khan, Z. Iqbal and W.K. Mashwani, "A survey on intrusion detection and prevention in wireless ad-hoc networks," *Journal of System Architecture*, vol.105, 101701, 2020.
- [6] K. Albulayhi et al., "An adaptive deep-ensemble anomaly-based intrusion detection system for the internet of things," In *Proceedings of the 2021 IEEE World AI IoT Congress (AllIoT)*, Seattle, WA, USA, pp.10-13, 2021.
- [7] H.H. Dang et al., "A PCA-based method for IoT network traffic anomaly detection", *International Conference on Advanced Communications Technology (ICACT)*, pp.381-386, 2018.
- [8] D. Kshirsagar et al., "A feature reduction based reflected and exploited DDoS attacks detection system," *Journal of Ambient Intelligence and Humanized Computing*, vol.13, pp.393-405, 2021.
- [9] N. Pushpa raj and K. Deepak, "Feature selection for intrusion detection system in Internet-of-Things (IoT)," *ICT Express* 7, vol. 2, pp.177-181, 2021.
- [10] C. Zhaomin et al., "Autoencoder-based network anomaly detection," *Wireless Telecommunications Symposium (WTS)*, pp.1-5, 2018.
- [11] Z. Yansen and L. Jinwei, "Research of network traffic anomaly detection model based on multilevel autoregression," *IEEE 7th International Conference on Computer Science and Network Technology (ICCSNT)*, pp.380-384, 2019.
- [12] A. Maryam and B. Keivan, "Using metaheuristic algorithms of genetic, particle swarm optimization and glow-worm in the intrusion detection system," *International Journal of Computer Science and Network Security*, vol.16, no.10, pp.78-86, 2016.
- [13] H.H. Ren et al., "An unsupervised deep learning model for early network traffic anomaly detection", *IEEE Access*, vol.8, pp.30387-30399, 2020.